

THE ECONOMICS OF ALTERNATIVE INVESTMENTS

Illiquidity Premium in Cryptocurrency Markets

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1. Introduction

Amihud (2002) establishes that illiquid equities command a return premium to compensate for trading costs. Whether this illiquidity premium extends to cryptocurrency markets remains an open empirical question. Using daily CoinMarketCap data for 23 cryptocurrencies (2013–2021), this paper tests for an illiquidity premium using portfolio sorts and Fama-MacBeth cross-sectional regressions.

2. Data and Construction

2.1 Data

The dataset includes daily Open-High-Low-Close prices, volume, and market capitalisation. The panel is unbalanced, as cryptocurrencies enter the sample at heterogeneous listing dates. The analysis excludes 640 zero-volume observations, pre-2014, to avoid distorted liquidity estimates. The final sample yields 36,442 coin-day observations, with minor variation across specifications.

2.2 Construction

The dependent variable is defined as

$$\ln(r)_{it} = \ln\left(\frac{Price_{it}}{Price_{i(t-1)}}\right), \quad (1)$$

where r_{it} represents daily log returns, derived from closing prices, for each cryptocurrency i at time t .

To mitigate measurement error and capture multiple dimensions of liquidity, this analysis employs alternative illiquidity measures. The Close-to-Close Amihud (2002) measure,

$$ILLIQ_{it} = \frac{|r_{it}|}{Volume_{it}}, \quad (2)$$

represents absolute returns with respect to trading volume. To refine this estimation of price impact, the High-to-Low Amihud measure is incorporated via the following estimator,

$$High - to - Low_{it} = \frac{\left| \frac{High_{it}}{Low_{it}} - 1 \right|}{Volume_{it}}. \quad (3)$$

As demonstrated by Lacava et al. (2026), this measure is more efficient than the Amihud (2002) proxy, as it utilises the absolute daily price range as an integrated volatility proxy.

Turnover, motivated by Datar et al. (1998), is employed to capture trading intensity through the relationship,

$$Turnover_{it} = \frac{Volume_{it}}{Market\ Capitalisation_{it}}. \quad (4)$$

While Barinov (2014) cautions that turnover may proxy for investor uncertainty rather than liquidity, the simultaneous use of the High-Low Amihud measure and turnover allows for the isolation of distinct dimensions of liquidity.

Liquidity measures are log-transformed to mitigate right-skewness. $ILLIQ_{it}$ and $High - to - Low_{it}$ use $\ln(x + 1)$ to accommodate zero values, while turnover uses $\ln(x)$. To address outliers, all continuous variables were winsorised at the 1st and 99th percentiles. Lagged market capitalisation and lagged log returns are included as controls for size and momentum, following Liu et al. (2022), who identify these as two of the three characteristics that price the cryptocurrency cross-section. The third, the market factor, is absorbed by the time-varying intercept in the Fama-MacBeth specification.

3. Preliminary Analysis

Table 1 and Figures 1 to 4 summarise the daily returns and three liquidity measures, revealing substantial cross-coin heterogeneity in both return volatility and liquidity.

Table 1: Summary Statistics

Variable	Mean	Standard deviation	Minimum value	Maximum value
$\ln(r)_{it}$	0.0018	0.0580	-0.1791	0.2112
$\ln(ILLIQ_{it} + 1)$	0.0643	0.2563	0	1.8900
$\ln(High - to - Low_{it} + 1)$	0.1199	0.4219	0	2.9000
$\ln(Turnover_{it})$	-2.8943	1.828	-7.2428	1.5559

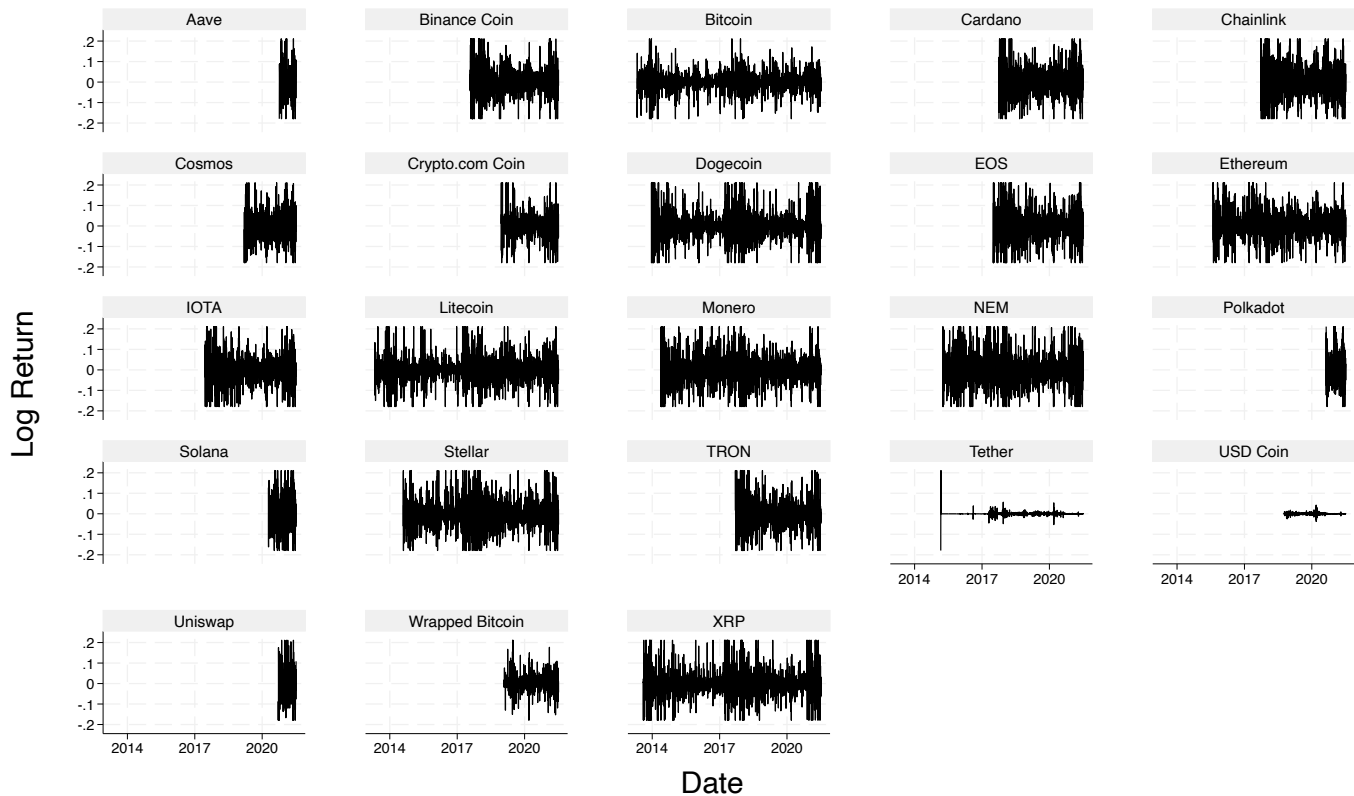


Figure 1: Daily Log Returns for All Cryptocurrencies

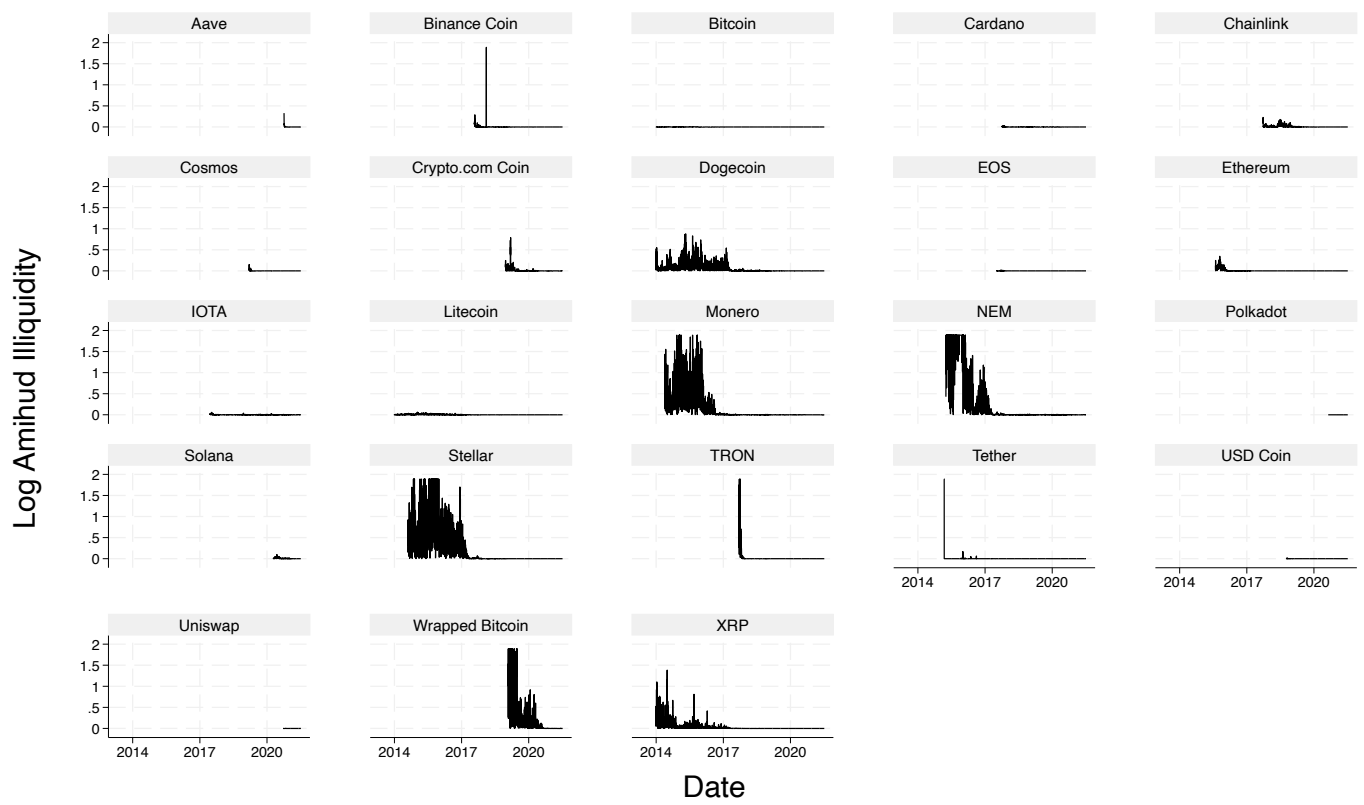


Figure 2: Log Amihud Illiquidity for All Cryptocurrencies

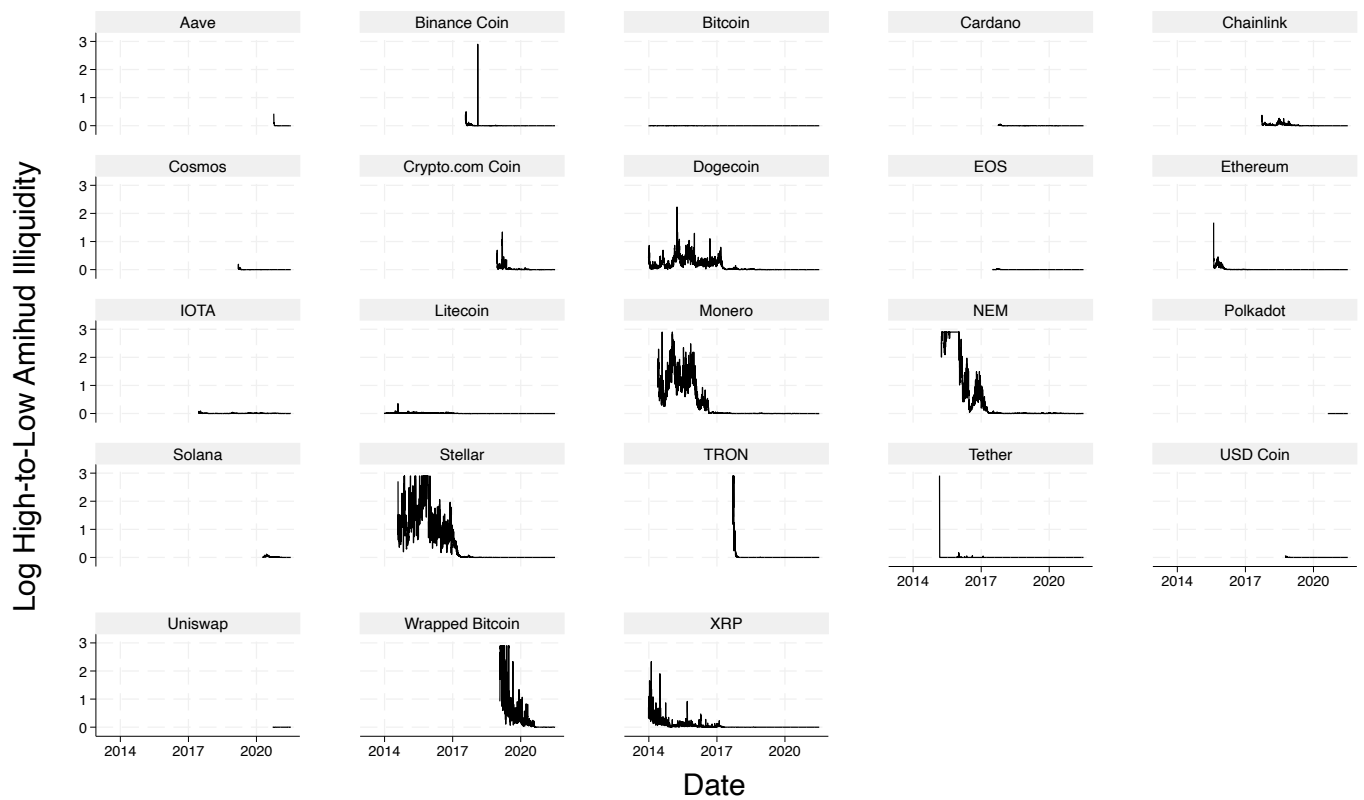


Figure 3: Log High-to-Low Amihud Illiquidity for All Cryptocurrencies

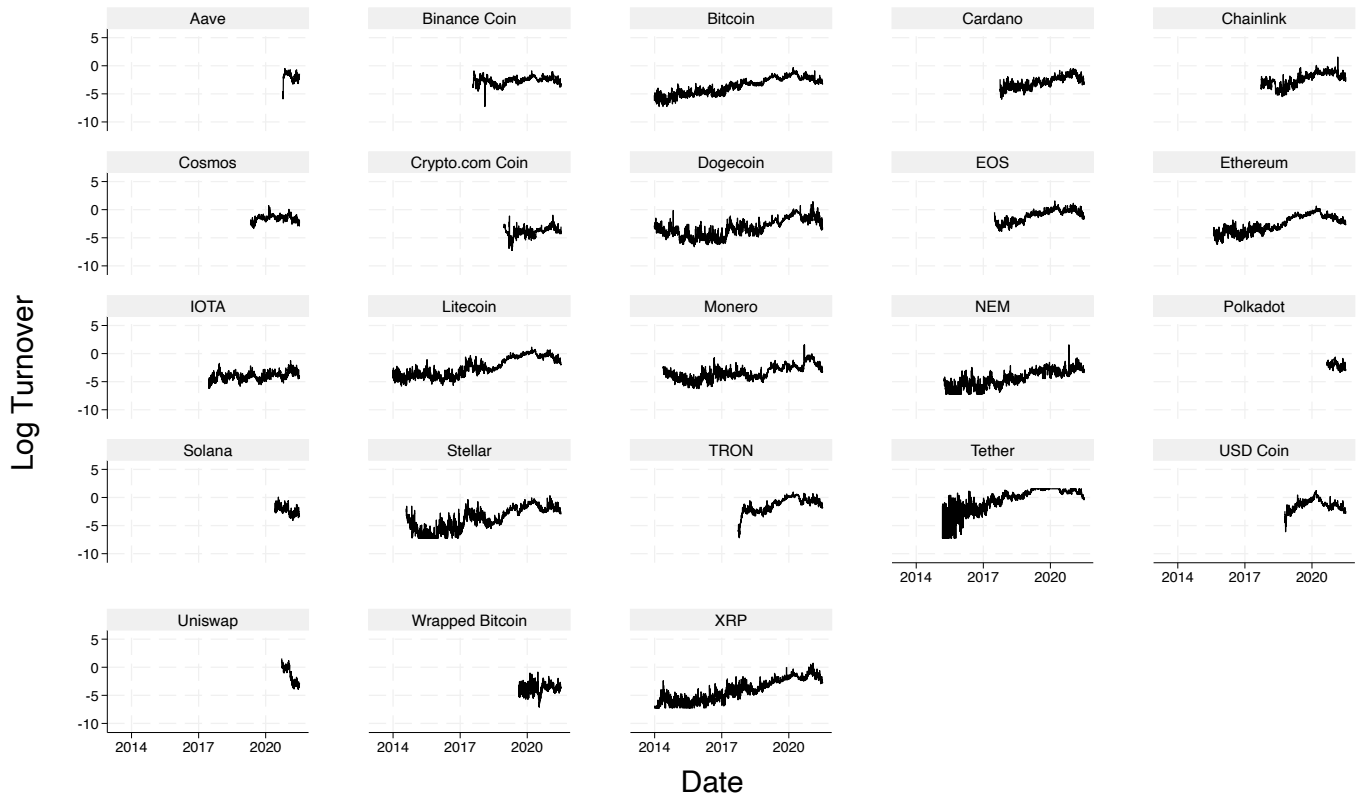


Figure 4: Log Turnover for All Cryptocurrencies

4. Illiquidity Premium

Two complementary methods test for an illiquidity premium. Portfolio sorts examine whether illiquid coins earn higher returns in the cross-section. Naïve Fama-MacBeth models then test this relationship in a regression framework, before controlled specifications account for size and momentum.

4.1 Portfolio Sorts

Coins are sorted each day into tercile portfolios based on the previous day's illiquidity, following Ali et al. (2025). Portfolio 1 contains the most liquid coins, Portfolio 3 the least liquid, and the illiquidity premium is defined as Portfolio 3 minus Portfolio 1. The one-day lag ensures that the sorting variable is observable before returns are realised. Significance of the premium is assessed using Newey-West (1987) standard errors with 8 lags following the bandwidth rule of Newey and West (1994),

$$Lags = 4\left(\frac{T}{100}\right)^{\frac{2}{9}}, \quad (5)$$

for $T = 2747$, to account for heteroskedasticity and autocorrelation in the daily premium series.

Table 2: Mean Daily Log Returns by Illiquidity Portfolio

	Portfolio 1	Portfolio 2	Portfolio 3	Premium
Log Amihud	0.0009	0.0018	0.0026	0.0015* (0.0008)
Log High-to-Low Amihud	0.0010	0.0018	0.0025	0.0012 (0.0007)
Log Turnover	0.0016	0.0020	0.0018	0.0000 (0.0007)

*Newey-West standard errors in parenthesis . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table 2 reports a positive and statistically significant illiquidity premium for the Amihud (2002) measure. The High-to-Low Amihud measure yields a positive but statistically insignificant premium. The turnover premium is statistically indistinguishable from zero ($p = 0.99$).

4.2 Fama-MacBeth

Fama-MacBeth (1973) regressions are employed to assess whether illiquidity is priced. The baseline models regress returns on illiquidity proxies in isolation. The core models then control for market capitalisation and lagged returns to account for size and momentum effects, following Liu et al. (2022). Robustness is assessed by substituting the price-impact measure with turnover. This three-stage approach isolates the liquidity premium while testing whether the results are robust to alternative proxies and additional controls.

The core specification estimated each day t across all coins is:

$$\ln(r)_{it} = \lambda_{0t} + \lambda_{1t}\ln(Liquidity)_{i(t-1)} + \lambda_{2t}\ln(Size)_{i(t-1)} + \lambda_{3t}\ln(r)_{i(t-1)} + \varepsilon_{it}, \quad (6)$$

where $\ln(Liquidity)_{i(t-1)}$ is the lagged log liquidity measure, $\ln(Size)_{i(t-1)}$ controls for market capitalisation, and $\ln(r)_{i(t-1)}$ controls for short-run momentum. The illiquidity premium is given by the time-series mean $\bar{\lambda}_1$, with inference based on Newey-West (1987) standard errors using 8 lags, for $T = 2,747$.

The time-varying intercept λ_{0t} absorbs market-wide shocks, providing implicit time fixed effects. Coin fixed effects are excluded since the approach exploits between-coin variation in characteristics.

Table 3: Fama-Macbeth Results

	Amihud	High-to-Low Amihud	Controlled Amihud	Controlled High-to-Low Amihud	Controlled Turnover
Log Amihud (lag)	−0.006 (0.768)		−0.025 (0.799)		
Log High-to-Low Amihud (lag)		0.143 (0.273)		0.009 (0.309)	
Log Turnover (lag)					−0.004 (0.003)
Log Market Capitalisation (lag)			−0.001 (0.001)	0.001 (0.000)	0.000 (0.000)
Log Return (lag)			−0.037 (0.066)	−0.021 (0.087)	−0.022 (0.117)
Constant	0.001 (0.001)	0.001 (0.001)	0.015 (0.017)	−0.015 (0.013)	−0.023 (0.021)
R^2	0.155	0.151	0.423	0.419	0.402
Observations	36,396	36,416	36,073	36,073	36,073

Newey-West standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The dependent variable is the daily log return.

Table 3 shows that no liquidity coefficient is statistically significant. The R^2 of approximately 0.42 falls to roughly 0.15 once adjusted for the small cross-section, indicating that the apparent fit largely reflects degrees-of-freedom mechanics rather than explanatory power. The regression evidence finds no robust illiquidity premium.

4.3 Interpretation of Empirical Findings

Portfolio sorts find a marginally significant Amihud illiquidity premium at the 10% level under Newey-West standard errors, consistent with Ali et al. (2025), while the High-to-Low Amihud and turnover measures are insignificant. The Fama-MacBeth regressions show no significant illiquidity effect across either the naïve or controlled specifications. This divergence is consistent with portfolio sorts capturing non-linear differences across the extreme terciles, while the Fama-MacBeth specification imposes linearity and uses controls that absorb part of the cross-sectional variation in illiquidity.

The analysis is based on an unbalanced panel of 23 cryptocurrencies, which restricts the inclusion of additional control variables and raises the risk of omitted variable bias from unobserved coin characteristics that influence both liquidity and returns. The results should therefore be interpreted as associational rather than causal.

Han (2022) suggests that liquidity risk, measured as sensitivity to aggregate liquidity shocks rather than the level of illiquidity, is significantly priced in cryptocurrency markets. This suggests that the null Fama-MacBeth results may reflect the dimensions of the liquidity proxies used, rather than the absence of an illiquidity premium.

5. Limitations of Daily Data

Estimating liquidity at daily frequency introduces measurement noise that is absent in high-frequency datasets. Lacava et al. (2026) demonstrate that the most precise approach is to compute realised illiquidity via Realised Power Variation. This study operates within the constraints of daily data, so it adopts the High-to-Low Amihud measure as the closest daily proxy, integrating the intraday price range as an integrated volatility proxy in place of the Realised Power Variation benchmark.

The use of low-frequency estimators in this market is supported empirically. Brauneis et al. (2021) compare daily proxies against high-frequency order book data across exchanges and show that the Amihud (2002) and High-to-Low Amihud measures track time-series variation in trading costs effectively. Goyenko et al. (2009) provide complementary cross-asset evidence that simple daily proxies recover the bulk of intraday spread variation. Lou and Shu (2017) supply the offsetting caveat. These measures capture the level of illiquidity, but their return predictability is often dominated by absolute returns and trading volume rather than a genuine liquidity premium, a risk that is pronounced in the cryptocurrency market where elevated idiosyncratic volatility may obscure the liquidity signal.

6. Relation to Alternative Assets

The regression results in this study indicate that the illiquidity premium in cryptocurrency is statistically insignificant. Liu and Tsyvinski (2021) document that cryptocurrency returns are uncorrelated with traditional macroeconomic and equity risk factors, suggesting that cross-sectional pricing in this market may be governed by characteristics specific to the asset class.

While Amihud (2002) identifies a significant illiquidity premium in equities, using Close-to-Close returns, this asset is inherently tied to traditional market hours. As argued by Barardehi et al. (2021), the Amihud proxy in stocks may suffer from measurement error because price jumps frequently occur during overnight periods without corresponding trading volume. In contrast, the continuous nature of cryptocurrency markets suggests a more direct relationship between volume and price response, providing a theoretically more accurate measure of transactional price impact.

Cryptocurrency lacks the capital lockups and statutory transaction costs that drive premia in other alternative assets. Private equity investors face extended capital lockups, which Franzoni et al. (2012) associate with an annual 3% liquidity risk premium. Institutional real estate carries high transaction costs and taxes, such as Stamp Duty Land Tax in the United Kingdom (Cheng et al., 2010), and Marcato (2018) documents a consistent 3.0% premium on this basis. Hwang et al. (2017), however, find that returns do not differ significantly between liquid and illiquid residential properties, indicating that cross-sectional variation in liquidity does not translate into a priced risk premium in every market.

While the ongoing tokenisation of real-world assets aims to enhance accessibility, Mafrur (2025) notes that valuation opacity and low trading volumes are likely to preserve a residual illiquidity premium in those markets. The null result documented here therefore appears specific to the structural features of cryptocurrency rather than evidence against the broader illiquidity-premium hypothesis.

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